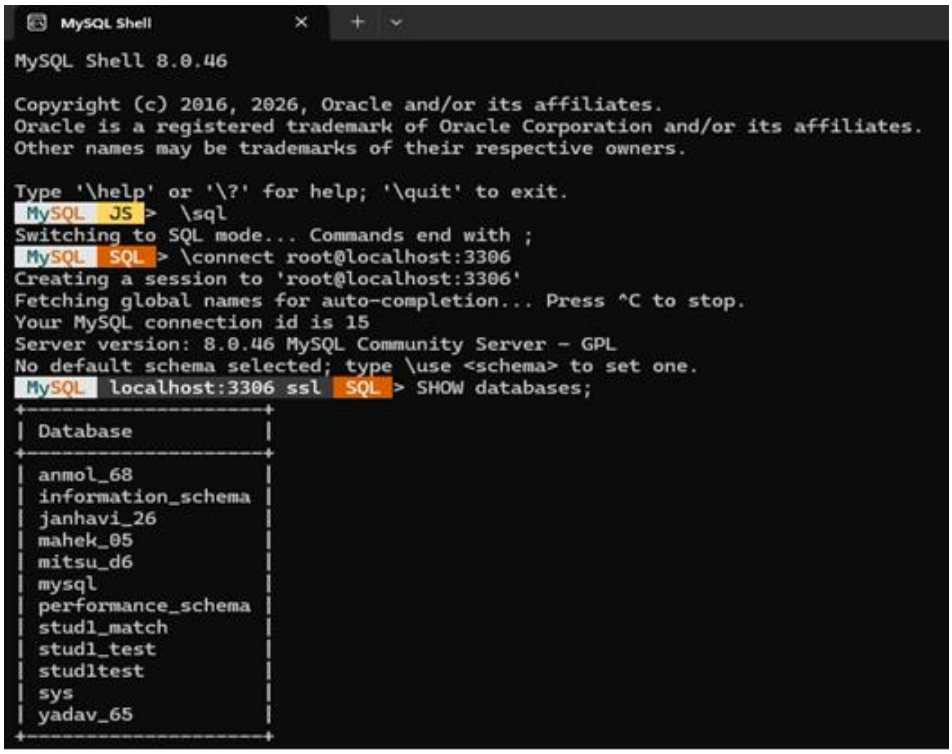
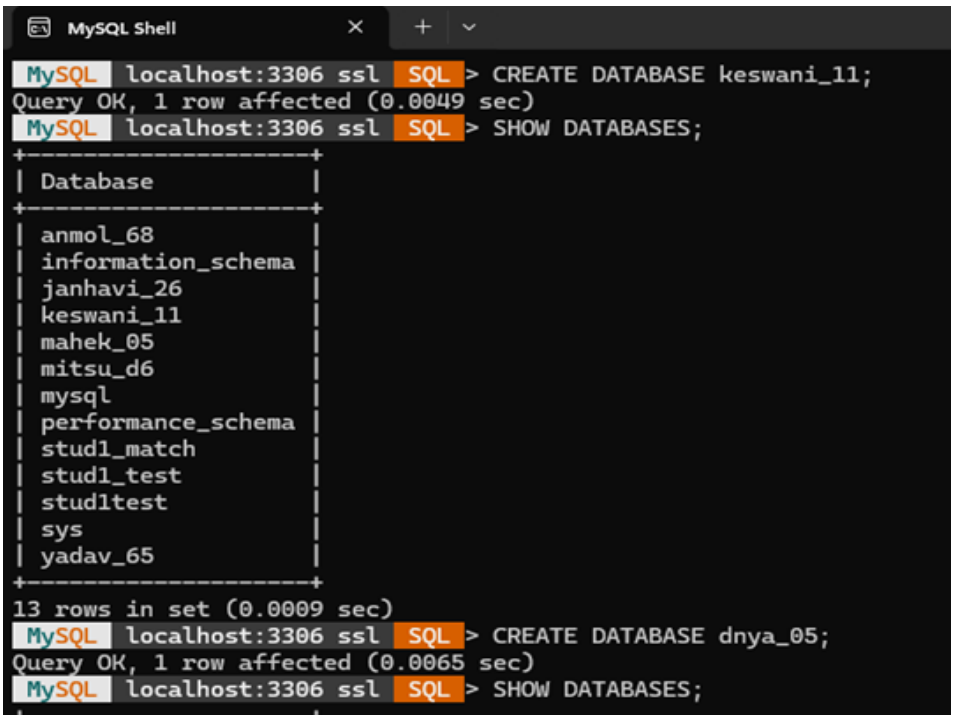
Lab 1

AIM: To perform and understand fundamental DBMS operations, including database creation , table management, data manipulation, modification, and retrieval using MySQL.





```
+--------------------+
| Database           |
+--------------------+
| anmol_68           |
| dnya_05            |
| information_schema |
| janhavi_26         |
| keswani_11         |
| mahek_05           |
| mitsu_d6           |
| mysql              |
| performance_schema |
| stud1_match        |
| stud1_test         |
| stud1test          |
| sys                |
| yadav_65           |
+--------------------+
14 rows in set (0.0010 sec)
MySQL localhost:3306 ssl SQL > DROP DATABASE mitsu_d6;
Query OK, 1 row affected (0.0119 sec)
```

```
MySQL localhost:3306 ssl SQL > SHOW databases;
+--------------------+
| Database           |
+--------------------+
| anmol_68           |
| dnya_05            |
| information_schema |
| janhavi_26         |
| keswani_11         |
| mahek_05           |
| mysql              |
| performance_schema |
| stud1_match        |
| stud1_test         |
| stud1test          |
| sys                |
| yadav_65           |
+--------------------+
13 rows in set (0.0010 sec)
MySQL localhost:3306 ssl SQL > USE yadav_65;
Default schema set to `yadav_65`.
Fetching global names, object names from `yadav_65` for auto-completion... Press ^C to stop.
MySQL localhost:3306 ssl yadav_65 SQL > CREATE TABLE Persons (
                                     -> PersonID int,
                                     -> LastName varchar(25),
                                     -> FirstName varchar(25),
                                     -> Address varchar(100),
                                     -> City varchar(25)
                                     -> );
Query OK, 0 rows affected (0.0101 sec)
MySQL localhost:3306 ssl yadav_65 SQL > SHOW TABLES;
+--------------------+
```

```
+---------------------+
| Tables_in_yadav_65 |
+---------------------+
| persons             |
+---------------------+
1 row in set (0.0011 sec)
MySQL localhost:3306 ssl yadav_65 SQL > DESCRIBE Persons;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| PersonID  | int         | YES  |     | NULL    |       |
| LastName  | varchar(25) | YES  |     | NULL    |       |
| FirstName | varchar(25) | YES  |     | NULL    |       |
| Address   | varchar(100)| YES  |     | NULL    |       |
| City      | varchar(25) | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
5 rows in set (0.0019 sec)
MySQL localhost:3306 ssl yadav_65 SQL > DROP TABLE Persons;
Query OK, 0 rows affected (0.0103 sec)
MySQL localhost:3306 ssl yadav_65 SQL > CREATE TABLE Customers (CustomerID int,LastName varchar(25),FirstName
                                     -> varchar(25),Address varchar(100),City varchar(25), primary key
                                     -> (CustomerID));
Query OK, 0 rows affected (0.0144 sec)
```

```
MySQL localhost:3306 ssl yadav_65 SQL > DESCRIBE Customers;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| CustomerID | int         | NO   | PRI | NULL    |       |
| LastName   | varchar(25) | YES  |     | NULL    |       |
| FirstName  | varchar(25) | YES  |     | NULL    |       |
| Address    | varchar(100)| YES  |     | NULL    |       |
| City       | varchar(25) | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
5 rows in set (0.0020 sec)
MySQL localhost:3306 ssl yadav_65 SQL > INSERT INTO Customers (CustomerID,LastName,FirstName,Address,City)
                                     -> VALUES (10,'yadav','khushi','A67,Building10,abc lane','Mumbai');
Query OK, 1 row affected (0.0048 sec)
MySQL localhost:3306 ssl yadav_65 SQL > SELECT * FROM Customers;
+------------+----------+-----------+---------------------+--------+
| CustomerID | LastName | FirstName | Address             | City   |
+------------+----------+-----------+---------------------+--------+
|         10 | yadav    | khushi    | A67,Building10,abc lane | Mumbai |
|      96365 | Shah     | Hiren     | A67,Building10,abc lane | Mumbai |
+------------+----------+-----------+---------------------+--------+
2 rows in set (0.0006 sec)
```

```
MySQL localhost:3306 ssl yadav_65 SQL > insert into customers values
                                     -> (300,'keswani','anmol','bk16 room 1','mumbai'),
                                     -> (400,'yadav','janhavi','jaoli apt','mumbai'),
                                     -> (500,'yadav','khushi','mayflower apt','pune'),
                                     -> (600,'punjabi','kani','mayflower','mumbai'),
                                     -> (700,'kalwani','palak','satnam apt','mumbai'),
                                     -> (800,'bhatnagar','mahek','tulip-b nisarg','mumbai'),
                                     -> (900,'sadwani','vivaan','lily-b nisarg','ulhasnagar'),
                                     -> (901,'tharyani','mayank','shiv pooja','ulhasnagar'),
                                     -> (902,'bhatnagar','ria','shiv pooja','mumbai'),
                                     -> (903,'mankani','dhiren','nisarg greens','mumbai');
Query OK, 10 rows affected (0.0043 sec)

Records: 10  Duplicates: 0  Warnings: 0
MySQL localhost:3306 ssl yadav_65 SQL > select *from customers;
+------------+-----------+-----------+--------------------------+-------------+
| CustomerID | LastName  | FirstName | Address                  | City        |
+------------+-----------+-----------+--------------------------+-------------+
|         10 | yadav     | khushi    | A67,Building10,abc lane  | Mumbai      |
|        300 | keswani   | anmol     | bk16 room 1              | mumbai      |
|        400 | yadav     | janhavi   | jaoli apt                | mumbai      |
|        500 | yadav     | khushi    | mayflower apt            | pune        |
|        600 | punjabi   | kani      | mayflower                | mumbai      |
|        700 | kalwani   | palak     | satnam apt               | mumbai      |
|        800 | bhatnagar | mahek     | tulip-b nisarg           | mumbai      |
|        900 | sadwani   | vivaan    | lily-b nisarg            | ulhasnagar  |
|        901 | tharyani  | mayank    | shiv pooja               | ulhasnagar  |
|        902 | bhatnagar | ria       | shiv pooja               | mumbai      |
|        903 | mankani   | dhiren    | nisarg greens            | mumbai      |
|      96365 | Shah      | Hiren     | A67,Building10,abc lane  | Mumbai      |
+------------+-----------+-----------+--------------------------+-------------+
12 rows in set (0.0006 sec)
```

```
MySQL localhost:3306 ssl yadav_65 SQL > alter table customers add emil varchar(50);
Query OK, 0 rows affected (0.0148 sec)

Records: 0  Duplicates: 0  Warnings: 0
MySQL localhost:3306 ssl yadav_65 SQL > describe customers;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| CustomerID | int         | NO   | PRI | NULL    |       |
| LastName   | varchar(25) | YES  |     | NULL    |       |
| FirstName  | varchar(25) | YES  |     | NULL    |       |
| Address    | varchar(100)| YES  |     | NULL    |       |
| City       | varchar(25) | YES  |     | NULL    |       |
| emil       | varchar(50) | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
6 rows in set (0.0016 sec)
MySQL localhost:3306 ssl yadav_65 SQL > alter table customers add remark varchar(255);
Query OK, 0 rows affected (0.0119 sec)

Records: 0  Duplicates: 0  Warnings: 0
MySQL localhost:3306 ssl yadav_65 SQL > describe customers;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| CustomerID | int         | NO   | PRI | NULL    |       |
| LastName   | varchar(25) | YES  |     | NULL    |       |
| FirstName  | varchar(25) | YES  |     | NULL    |       |
| Address    | varchar(100)| YES  |     | NULL    |       |
| City       | varchar(25) | YES  |     | NULL    |       |
| emil       | varchar(50) | YES  |     | NULL    |       |
| remark     | varchar(255)| YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
7 rows in set (0.0015 sec)
```

```
MySQL localhost:3306 ssl yadav_65 SQL > select *from customers;
+------------+-----------+-----------+-----------------------+-------------+------+---------+
| CustomerID | LastName  | FirstName | Address               | City        | emil | remark  |
+------------+-----------+-----------+-----------------------+-------------+------+---------+
|         10 | yadav     | khushi    | A67,Building10,abc lane | Mumbai    | NULL | NULL    |
|        300 | keswani   | anmol     | bk16 room 1           | mumbai      | NULL | NULL    |
|        400 | yadav     | janhavi   | jaoli apt             | mumbai      | NULL | NULL    |
|        500 | yadav     | khushi    | mayflower apt         | pune        | NULL | NULL    |
|        600 | punjabi   | kani      | mayflower             | mumbai      | NULL | NULL    |
|        700 | kalwani   | palak     | satnam apt            | mumbai      | NULL | NULL    |
|        800 | bhatnagar | mahek     | tulip-b nisarg        | mumbai      | NULL | NULL    |
|        900 | sadwani   | vivaan    | lily-b nisarg         | ulhasnagar  | NULL | NULL    |
|        901 | tharyani  | mayank    | shiv pooja            | ulhasnagar  | NULL | NULL    |
|        902 | bhatnagar | ria       | shiv pooja            | mumbai      | NULL | NULL    |
|        903 | mankani   | dhiren    | nisarg greens         | mumbai      | NULL | NULL    |
|      96365 | Shah      | Hiren     | A67,Building10,abc lane | Mumbai    | NULL | NULL    |
+------------+-----------+-----------+-----------------------+-------------+------+---------+
12 rows in set (0.0006 sec)
```

```
MySQL localhost:3306 ssl yadav_65 SQL > update customers
                                    -> set emil='ramesh.joshi@abc.com',
                                    -> remark='regular buyer'
                                    -> where customerID=89790;
Query OK, 0 rows affected (0.0083 sec)

Rows matched: 0  Changed: 0  Warnings: 0
MySQL localhost:3306 ssl yadav_65 SQL > SELECT CustomerID,Address FROM Customers;
+------------+-------------------------+
| CustomerID | Address                 |
+------------+-------------------------+
|         10 | A67,Building10,abc lane |
|        300 | bk16 room 1             |
|        400 | jaoli apt               |
|        500 | mayflower apt           |
|        600 | mayflower               |
|        700 | satnam apt              |
|        800 | tulip-b nisarg          |
|        900 | lily-b nisarg           |
|        901 | shiv pooja              |
|        902 | shiv pooja              |
|        903 | nisarg greens           |
|      96365 | A67,Building10,abc lane |
+------------+-------------------------+
12 rows in set (0.0006 sec)
MySQL localhost:3306 ssl yadav_65 SQL >
```

1. What is a Database? Why do we need a Database Management System such as MySQL?

A database is a collection of related information that is stored in an organized way. It makes it easier to store, find, update, and manage large amounts of data.

A Database Management System (DBMS) is software that helps users create and manage databases. MySQL is a widely used relational DBMS that uses SQL to perform different operations on data.

A DBMS such as MySQL is needed because it:

- Organizes data in a proper and systematic manner.
- Makes searching and retrieving information easier.
- Allows users to add, change, and remove data.
- Provides security by controlling who can access the data.
- Helps maintain accurate and consistent information.
- Can manage large amounts of data efficiently.
- Allows multiple users to access and work with the database.

Thus, a database is used to store information, while a DBMS such as MySQL provides the necessary tools to manage that information effectively.

2. What is the Difference between CREATE DATABASE, SHOW DATABASES, and DROP DATABASE?

- CREATE DATABASE — This command is used to make a new database in MySQL. It creates a new database where tables and related information can be stored.

Example:

CREATE DATABASE college;

- SHOW DATABASES — This command displays all the databases that are currently available on the MySQL server.

Example:

SHOW DATABASES;

- DROP DATABASE — This command is used to remove an existing database completely. It also deletes all the tables and data contained in that database.

Example:

DROP DATABASE college;

In simple terms, CREATE DATABASE creates a database, SHOW DATABASES displays available databases, and DROP DATABASE deletes a database.

3. What is a Table in a Database? Explain the Purpose of Rows and Columns.

A table is a basic structure used in a database to store related information. Data in a table is arranged using rows and columns, which makes the information easy to understand and manage.

- Rows — A row contains the complete details of one particular record. For example, in a student table, one row may contain the ID, name, age, and department of one student.
- Columns — A column represents one particular type of information or attribute. For example, a student table may have columns such as StudentID, Name, Age, and Department.

Therefore, a row represents one complete record, while a column represents a particular field or attribute of the records.

4. What is a Primary Key? Why was CustomerID selected as the primary key in the Customers table?

A primary key is a column, or a combination of columns, that uniquely identifies every record in a table. The values in a primary key must be unique, and a primary key cannot contain NULL values.

CustomerID is selected as the primary key in the Customers table because every customer can be given a different identification number. This makes it easy to distinguish one customer from another, even if two customers have the same name or other similar details.

For example, if CustomerID 101 belongs to one customer, no other customer should have CustomerID 101.

Therefore, CustomerID is appropriate as a primary key because it provides a unique identification for every customer and helps maintain the accuracy of the table.

5. What is the purpose of the VARCHAR and INT data types? Give one suitable example of each.

- VARCHAR — VARCHAR is used to store text or character values whose length can vary. It can be used for names, addresses, email IDs, and other text-based information.

Example:

CustomerName VARCHAR(50)

This can store a customer's name with a maximum length of 50 characters.

- INT — INT is used to store whole numbers without decimal values. It is commonly used for IDs, ages, quantities, marks, and counts.

Example:

CustomerID INT

This can store the numerical ID assigned to a customer.

Thus, VARCHAR is mainly used for text values, whereas INT is used for whole-number values.

6. What is the Difference between INSERT, SELECT, UPDATE, and ALTER TABLE?

- INSERT — The INSERT command is used to add new records to an existing table.

Example:

INSERT INTO Customers VALUES (101, 'Rahul', 'Mumbai');

- SELECT — The SELECT command is used to retrieve and display information stored in a table.

Example:

SELECT * FROM Customers;

- UPDATE — The UPDATE command is used to change existing information in one or more records.

Example:

UPDATE Customers

SET City = 'Pune'

WHERE CustomerID = 101;

- ALTER TABLE — The ALTER TABLE command is used to change the structure of an existing table. For example, it can be used to add or remove columns.

Example:

ALTER TABLE Customers ADD Email VARCHAR(50);

In short, INSERT adds new records, SELECT displays data, UPDATE changes existing data, and ALTER TABLE changes the structure of a table.

7. Why is the WHERE clause important in an UPDATE statement?

The WHERE clause is important because it tells MySQL which specific records should be changed. It prevents the UPDATE command from modifying records that are not supposed to be changed.

For example:

```
UPDATE Customers
SET City = 'Pune'
WHERE CustomerID = 101;
```

In this example, only the customer whose CustomerID is 101 will have their city changed to Pune.

If the WHERE clause is not used:

```
UPDATE Customers
SET City = 'Pune';
```

the city will be changed to Pune for every customer in the table.

Therefore, using the WHERE clause with UPDATE helps make sure that only the required records are modified and prevents accidental changes to the entire table.